PanNET patient data collection tables

Table 1. Variables 1 to 7: demographic and pathological information

Variable	Variable name as it appears in SEER	Description	Permitted source values
1. Age at diagnosis	Age recode with single ages and 90+	Age in completed years when the tumor was diagnosed.	Single-year labels, e.g. 55 years; 90+ years (coded as 90).
2. Race	Race recode (W, B, AI, API)	Recorded race category.	White; Black; Asian or Pacific Islander; American Indian/Alaska Native; Unknown.
3. Sex	Sex	Sex recorded in the source dataset.	Female; Male.
4. Marital status at diagnosis	Marital status at diagnosis	Recorded marital status at diagnosis.	Married (including common law); Single (never married); Divorced; Widowed; Separated; Unmarried or Domestic Partner; Unknown.
5. Histological diagnosis	ICD-O-3 Hist/behav, malignant	Tumor type defined by ICD-O-3 morphology and behavior. /3 denotes malignant behavior.	Code plus diagnostic label. Accepted historical codes and labels: Table 7.
6. Histological grade	Grade Recode (thru 2017)	Historical registry grade; distinct from T/N stage and modern WHO NET grade.	Grade I, II, III, IV; Unknown. Full exported labels: Table 9.
7. Primary pancreatic site	Primary Site - labeled	Anatomical site of the pancreatic primary tumor.	ICD-O topography code plus site label: Table 8.

Stage, treatment, size and metastases

Table 2. Variables 8 to 17: stage and other clinical information

Variable	Variable name as it appears in SEER	Description	Permitted source values
8. Primary tumor extent (T)	Derived AJCC T, 7th ed (2010-2015) Derived SEER Combined T (2016-2017)	Size/local extent of the primary. 2010-2015 uses AJCC 7th; 2016 uses UICC 7th-based SEER Combined.	T0-T4; TX = unassessable. 2016 uses c/p prefixes (e.g. c1/p1). Definitions: Table 4.
9. Regional node status (N)	Derived AJCC N, 7th ed (2010-2015) Derived SEER Combined N (2016-2017)	Absence or presence of regional nodal metastasis under the corresponding seventh-edition framework.	N0; N1; NX = unassessable. 2016 uses c/p prefixes (e.g. c0/p0). Definitions: Table 5.
10. Primary tumor surgery	RX Summ--Surg Prim Site (1998+)	Whether pancreatic primary-site surgery was performed in the initial treatment course.	No = 0/00; Yes = observed codes 25, 30, 35, 36, 37, 40, 60, 70, 80, 90; Unknown = 99.
11. Regional node procedure	RX Summ--Scope Reg LN Sur (2003+)	Regional node biopsy/removal procedure. Separate from nodal status or positive-node count.	None; biopsy, sentinel or removal categories; Unknown or not applicable. Exact labels: Table 11.
12. Radiotherapy	Radiation recode	Recorded radiation treatment in the initial treatment course.	Radiation modality; None/Unknown; Refused; Recommended, unknown if administered. Exact labels: Table 12.
13. Chemotherapy	Chemotherapy recode (yes, no/unk)	Recorded chemotherapy in the initial treatment course.	Yes; No/Unknown. The latter is one combined SEER category, not proof that no treatment occurred.
14. Primary tumor size	Tumor Size Over Time Recode (1988+)	Largest primary tumor dimension in millimetres.	Numeric mm; 0 only for no demonstrable primary mass; Unknown or size unreasonable. Later size groups: Table 10.
15. Bone metastasis	SEER Combined Mets at DX-bone (2010+)	Bone metastasis present at initial diagnosis.	Yes; No; Unknown.
16. Brain metastasis	SEER Combined Mets at DX-brain (2010+)	Brain metastasis present at initial diagnosis.	Yes; No; Unknown.
17. Lung metastasis	SEER Combined Mets at DX-lung (2010+)	Lung metastasis present at initial diagnosis.	Yes; No; Unknown.

Model outcomes and stage definitions

Table 3. Additional variables needed to evaluate models

Variable	Variable name as it appears in SEER	Description	Permitted source values
Liver metastasis at diagnosis	SEER Combined Mets at DX-liver (2010+)	Observed liver-metastasis target; also identifies the liver-metastatic survival cohort.	Yes; No; Unknown. Unknown cannot provide an evaluation label.
Survival / follow-up time	Survival months	Overall-survival outcome time, from diagnosis to death or last follow-up.	Nonnegative numeric months; zero-month records may be valid.
Vital status	Vital status recode (study cutoff used)	Overall-survival event or censoring status. All-cause death is used.	Alive = censored (0); Dead = event (1).

Table 4. T categories: AJCC 7th edition and original ENETS [1, 2]

Category	AJCC 7th pancreatic definition	Original ENETS pancreatic definition (2006)
T0	No demonstrable primary tumor.	No demonstrable primary tumor.
T1	Confined to pancreas; size ≤20 mm.	Confined to pancreas; size <20 mm.
T2	Confined to pancreas; size >20 mm, with no upper size threshold.	Confined to pancreas; size 20-40 mm.
T3	Extension beyond pancreas without celiac-axis or superior-mesenteric-artery involvement.	Confined tumor >40 mm, or invasion of duodenum/bile duct.
T4	Involvement of celiac axis or superior mesenteric artery.	Invasion of adjacent organs or major vessel walls.
TX	Primary tumor cannot be assessed; retain as unknown.	Primary tumor cannot be assessed.

Table 5. N categories: regional lymph nodes [1, 2]

Category	AJCC 7th / original ENETS meaning
N0	No regional nodal metastasis.
N1	Regional nodal metastasis present.
NX	Regional nodal status cannot be assessed.

Table 6. M categories: stage reference only [1, 2]

Category	Shared basic meaning
M0	No distant metastasis identified.
M1	Distant metastasis present.
Unknown	Distant spread not adequately assessed.

Histological diagnosis and pancreatic site

Table 7. ICD-O-3 morphology and behavior values

Code	Historical diagnostic label
8150/3	Pancreatic endocrine tumor, malignant
8151/3	Insulinoma, malignant
8152/3	Glucagonoma, malignant
8153/3	Gastrinoma, malignant
8155/3	Vipoma, malignant
8156/3	Somatostatinoma, malignant
8240/3	Carcinoid tumor, NOS
8242/3	Enterochromaffin-like cell tumor, malignant
8243/3	Goblet cell carcinoid
8246/3	Neuroendocrine carcinoma, NOS
8249/3	Atypical carcinoid tumor

Table 8. ICD-O topography values: primary pancreatic site [4]

Code	Source site label / anatomical meaning	Later grouping
C25.0	Head of pancreas	Head
C25.1	Body of pancreas	Body
C25.2	Tail of pancreas	Tail
C25.3	Pancreatic duct	Others
C25.4	Islets of Langerhans	Others
C25.7	Other specified parts of pancreas	Others
C25.8	Overlapping lesion of pancreas	Others
C25.9	Pancreas, NOS / unspecified	Others

Table 9. Historical SEER grade values [3]

Value	Exported label
I / 1	Well differentiated; Grade I
II / 2	Moderately differentiated; Grade II
III / 3	Poorly differentiated; Grade III
IV / 4	Undifferentiated; anaplastic; Grade IV
Unknown	Unknown

Size and treatment values

Table 10. Primary tumor size: permitted measurements and later groups

Exact size in mm	Later group	Meaning
0 ≤ size ≤26	0-26 mm	Includes 26 mm. Zero only if no primary mass is demonstrable.
26 < size ≤36	26-36 mm	Includes 36 mm; excludes 26 mm.
Size >36	>36 mm	Exact measurement retained before grouping.
Unknown / size unreasonable	Missing	No group; do not substitute zero.

Table 11. Regional lymph-node procedure: exact SEER values

Source value	Later binary value
None	No / 0
Biopsy or aspiration of regional lymph node, NOS; Sentinel lymph node biopsy; Number of regional lymph nodes removed unknown; 1 to 3 regional lymph nodes removed; 4 or more regional lymph nodes removed; Sentinel node biopsy and lym nd removed same/unstated time	Yes / 1 for each listed procedure category
Unknown or not applicable	Missing

Table 12. Radiotherapy: exact SEER values

Source value	Later binary value
Beam radiation; Radioactive implants (includes brachytherapy) (1988+); Radioisotopes (1988+); Combination of beam with implants or isotopes; Radiation, NOS method or source not specified	Yes / 1
None/Unknown; Refused (1988+)	No / 0
Recommended, unknown if administered	Missing

Table 13. Sources and references

Reference	Source
[1] AJCC / UICC	AJCC 7th Manual, pancreas chapter, printed p. 245; SEER Combined staging basis (2016-2017).
[2] Original ENETS	Rindi et al. (2006), pancreatic TNM, Table 3.
[3] Historical grade	SEER Inquiry 20160023: legacy NET grade rules.
[4] Site codes	SEER Training: pancreatic ICD-O site codes.
[5] Current AJCC	AJCC current staging systems (2026); Version 9 overview.
[6] PanNET protocol	CAP endocrine pancreas v5.0.1.0 (June 2024), AJCC Version 9.
[7] Current ENETS	Current guidance index; 2023 nonfunctioning PanNET and functioning syndromes.

AJCC 7th edition and current frameworks

Current status (verified 8 October 2026). Pancreatic NETs use **AJCC Version 9**, effective from January 2024. AJCC updates are specific to each cancer site. ENETS currently lists its **2023 pancreatic guidance papers**; these are clinical guidance, rather than a numbered staging edition corresponding to AJCC Version 9. [5, 7]

Table 14. Main differences and similarities for pancreatic NETs [1, 2, 5-7]

Aspect	AJCC 7th / project framework	Current AJCC Version 9	ENETS pancreatic framework
Primary tumor size (T)	T1: confined, ≤20 mm. T2: confined, >20 mm, without an upper size boundary.	Confined tumors: T1 ≤20 mm; T2 >20 to ≤40 mm; T3 >40 mm.	Size-based 20/40 mm framework. Original ENETS definitions: T1 <20; T2 20-40; T3 >40 mm.
Local invasion (T)	T3: extension beyond pancreas without celiac-axis/superior-mesenteric-artery involvement. T4: involvement of those arteries.	Duodenum/ampulla/bile duct invasion: T3. Adjacent organs or major vessel walls: T4. Peripancreatic fat alone does not upstage.	Duodenum/bile duct invasion: T3. Adjacent organs or major vessel walls: T4. Current guidance describes this broadly similar T3/T4 approach.
Regional nodes (N)	N0: absent; N1: present. The project has a separate node-procedure flag.	N0/N1 retained; no N2 category for PanNET. Node counts can be reported separately.	Same basic N0/N1 distinction. Nodal involvement and node procedures/counts remain separate concepts.
Distant metastases (M)	M0/M1. The models use separate site flags, including liver status, rather than M as a predictor.	M0/M1, with M1a (liver only), M1b (extrahepatic only), M1c (both).	M0/M1 indicates distant-spread status. Clinical evaluation also distinguishes the involved sites, especially liver involvement.
Histology and grade	Legacy ICD-O labels and SEER grade I-IV. Original grading basis is not available in this export.	WHO NET G1-G3 based on Ki-67 and mitoses per 2 mm2; well-differentiated NET G3 is distinguished from poorly differentiated NEC.	Current guidance uses proliferation-based WHO NET grading (G1-G3), with differentiation considered separately. This does not directly replace legacy SEER grade I-IV.
Scope of the NET framework	Historical pancreatic staging and the broad code list in Table 7.	Dedicated well-differentiated PanNET framework. Poorly differentiated pancreatic NEC uses the pancreatic carcinoma framework.	2023 papers address well-differentiated nonfunctioning PanNET and functioning syndromes; digestive NEC has separate guidance.

Example: a 50 mm tumor confined to the pancreas is **AJCC 7th T2**, but **AJCC Version 9 / ENETS T3**. At exactly 20 mm, AJCC Version 9 specifies T1, whereas the original ENETS proposal specifies T2. Use the stated protocol for boundary cases. [1, 2, 6]

For this project: preserve the reported system/edition and underlying size/invasion findings when interpreting a current stage. A T number or modern WHO grade alone is not a reliable replacement for the historical model input. This section compares current practice with the 7th-edition framework; it does not attribute every difference to the Version 9 update.